S5 Table . Data extraction form, including primary outcomes and risk of bias questions

Section	Question	Answers
Study Characteristics	What year was the article published?	
	Where were the beaches located? (Region/city, province/state, country)	
	Number of beaches included in analysis	
	Number of swimming seasons' data used in model building	
Analytical Methods	Type of statistical model(s) used in final analysis	○ Multilinear regression ○ Bayesian networks ○ Artificial neural networks ○ Tree regression and/or random forests ○ Other: ____
	Possible explanatory variables assessed	○ Rainfall <24 hr ○ Rainfall 24hr ○ Rainfall 48 hr ○ Rainfall 72+ hr ○ Temperature ○ Wave height ○ River/stream outflow of water ○ Sewer outflow [FIB] ○ Previous day [FIB] ○ Solar radiation ○ Barometric pressure ○ Turbidity ○ Wind speed ○ Wind direction ○ Relative humidity ○ Discharge/flow (m3/s) ○ Conductivity ○ pH ○ Chlorophyll a ○ Total nitrogen ○ Total phosphorus ○ Other:_____
	Explanatory variables used in final model	
	Laboratory testing used, type of bacterial indicator, and how the bacterial concentration is reported	○ E. coli ○ Enterococcus ○ Other:_____ ○ Categorical ○ Continuous ○ Log transformed
	Performance measurements	Sensitivity = Specificity = R^2 =

		AUC (c-statistic) = Other performance measures: __
	Has this model been used in practice?	○ Yes, with data publicly available ○ Yes, with data kept internally ○ No ○ Unsure
	What was the method of measuring predictors/ source of predictors?	○ Governmental data ○ Collected by beach management ○ Conservation Authorities ○ Measured by researchers ○ Unsure
	Type of predictors in model	○ Continuous ○ Categorical ○ Transformed If transformed, reason for transforming: _____
	Handling of missing data	Handling of missing data: ___
	Are the modelling assumptions satisfied?	○ Yes ○ No ○ Unsure
	Were predictor weights adjusted to address overfitting of model?	○ Yes ○ No ○ Not applicable ○ Unsure
	How was the model validated?	○ Bootstrapping ○ Division of original dataset ○ Temporal validation ○ Geographical validation
	Were the distributions of predictors compared for development and validation datasets?	○ Yes ○ No ○ Unsure
	Predictor selection method (including preselection methods if relevant)	○ Full model approach ○ Backward elimination ○ Forward selection ○ Akaike Information Criterion ○ Bayesian Information Criterion ○ C-index ○ Nominal p-value ○ Other: _____